

Inferred solution of Unsteady Stokes Flow using Physics Informed Gaussian Process

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1 Introduction

The full posterior can be expressed as following, where $\mathcal{G}^{t-1}$ is defined as the *RHS* of the unsteady stokes equation.

$$\begin{bmatrix} u^t \\ p^t \end{bmatrix} \mid \begin{bmatrix} u_b \\ S_L^1 \\ S_p^1 \\ \mathcal{G}^{t-1} \\ s \end{bmatrix} \sim \mathcal{N}(\mathbf{q}^{\mathbf{T}})$$

Subscripts in math mode are written as a_b and superscripts are written as a^b . These can be combined and nested to write expressions such as

$$T_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = T(x^{i_1}, \dots, x^{i_p}, e_{j_1}, \dots, e_{j_q})$$

We write integrals using $\int$ and fractions using $\frac{a}{b}$. Limits are placed on integrals using superscripts and subscripts:

$$\int_0^1 \frac{dx}{e^x} = \frac{e-1}{e}$$

Lower case Greek letters are written as ω δ etc. while upper case Greek letters are written as Ω Δ . This is Nabla, ∇ , and Laplace

Mathematical operators are prefixed with a backslash as $\sin(\beta)$, $\cos(\alpha)$, $\log(x)$ etc.

In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Note' right now the optimizer is optimizing the same function on the same value. At the start of each loop, the data should reset so a new hyperparam could be train on new data.